



Harshitha S 250701234-2025-CSE ▾

H2

Started on	Friday, 21 August 2026, 4:08 PM
State	Finished
Completed on	Friday, 21 August 2026, 4:09 PM
Time taken	45 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, x;
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     scanf("%d", &x);
14
15     int low = 0, high = n - 1;
16     int floor = -1;
17
18     while (low <= high)
19     {
20         int mid = (low + high) / 2;
21
22         if (a[mid] == x)
23         {
24             floor = a[mid];
25             break;
26         }
27         else if (a[mid] < x)
28         {
29             floor = a[mid];
30             low = mid + 1;
31         }
32         else
33         {
34             high = mid - 1;
35         }
36     }
37
38     printf("%d\n", floor);
39
40     return 0;
41 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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